Modified peptides

How many distinct modified sequences carrying each of three tracked modifications were detected in each sample.

AT A GLANCE

Where	Peptide Level → Modification Plot
Needs	A peptide report, loaded as the Peptide level in Data → Upload, and an annotation. The peptide and precursor levels are read from the long-format report on disk rather than from a wide matrix, and this page needs an abundance in it: the report must carry the column Precursor.Quantity, which is fixed in the code and is not the **Abundance Column** offered on the quality-control pages. It also needs Modified.Sequence.
Annotation	With a protein dataset loaded, the protein annotation is used. Without one, the peptide annotation made in Data → Annotation is. Until one of them exists the page says so and draws nothing.
Recognises	Three classes, matched by name or by UniMod identifier: Carbamidomethylation (UniMod:4), Oxidation (UniMod:35) and Acetylation (UniMod:1). Nothing else is counted.
One bar	One sample and one modification. Its height is a count of distinct *modified sequences*, not of precursors and not of occurrences.
Table	None.

1 What the plot answers

Use these counts to compare the tracked modification classes between samples. Differences can reflect preparation, storage, search settings, identification depth or biology. Acetylation and oxidation are not automatically preparation artefacts. These are counts of detected sequences, not modification occupancy: a high bar does not say what fraction of all molecules carries a modification.

The bars are grouped by sample, so the comparison the figure invites is across samples within one modification, and that is the comparison it supports. Comparing the three modifications with each other on one sample compares three different things: how often a residue was chemically derivatised on purpose, how often one was oxidised, and how often one was acetylated.

> **READ COUNTS AS COUNTS**
>
> Suppose one sample has 100 distinct sequences carrying oxidation and another has 200. The second bar is twice as high, but this is not evidence of twice the oxidation rate: the second sample may also have twice as many identified sequences. Read modification counts alongside coverage and the search settings that determined which modifications were reported.

Reading it

- One bar: one modification in one sample. Its height is how many distinct modified sequences carrying that modification were detected there.
- The group: the three bars over one sample, drawn side by side at a fixed width and centred on the sample's tick.
- The label under each group: `condition_n`, numbered within the condition rather than across the experiment, in annotation order. With **Toggle IDs** on, the sample's own identifier is added on a second line.
- The legend, titled Modification. Only the classes with at least one positive count anywhere in the dataset appear in it; a class detected nowhere is drawn nowhere and named nowhere.

> **EASY TO GET WRONG**
>
> A sequence contributes at most once to a class per sample, however many times it carries that modification, and a sequence carrying two of the three tracked classes contributes once to each. So the bar is a count of sequences, not of modification sites and not of spectra — a peptide with two oxidised methionines is one, and a peptide seen at three charges in one run is also one.

2 Controls

Controls on this page

Control	What it does	Default	Where
Toggle IDs	Adds the sample's own identifier under its condition_n label, on a second line.	off	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm and DPI between 72 and 600, and a very large figure has its resolution reduced so that the image stays under 24 megapixels.	25 cm × 15 cm, 100	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Legend	Shown or hidden, and its title, size, colour, frame and position. Its entries are the modifications.	shown	appearance
Title, X axis, Y axis	Text, size, weight, colour and position of each label, and the size, rotation and colour of the tick labels. The plot writes Modifications per sample, Sample and Number of modified peptides, and rotates the sample labels by 90°.	the plot's own	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Frame and grid	Which of the four spines are drawn, and the grid behind the bars.	all four spines	appearance
Background	Paper colour, transparency and the preset themes.	default	appearance
Value labels	Offered on this plot, but the figure prints no numbers on its bars for it to size.	8	appearance

> **NOTE**
>
> There is no bar-colour control here. The three colours are fixed in the code — carbamidomethylation blue, oxidation red, acetylation green — and the appearance panel offers no colour card for this plot, so the same class is the same colour in every figure and in the summary report.

The panel offers no **Value labels** card on this page, because these bars carry no printed counts.

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

HOW IT IS COMPUTED

The long report is pivoted into a wide matrix: one row per distinct Modified.Sequence, one column per value of the file-name column, and in each cell the *largest* Precursor.Quantity of the rows that share that sequence and that run. Several precursors of one sequence in one run therefore collapse into one number before anything is counted.

The columns are then matched to the annotation and renamed condition_n. An exact run name is matched first; only a metadata sample that names no column exactly falls back to matching on the trailing number in the name. Columns the annotation does not account for are dropped, and the samples are drawn in annotation order.

A sequence counts as *detected* in a sample when its cell holds a finite number other than zero. Zero is read as not detected here, not as a measured zero.

For each tracked class m and each sample j the bar is then

$$\text{count}_{j,m} \;=\; \sum_{s \in S} \mathbf{1}[\, m \text{ matches } s \,] \cdot \mathbf{1}[\, s \text{ detected in } j \,],$$

where S is the set of distinct modified sequences in the report. The indicator on the left is a property of the sequence string alone, so it is the same in every sample; the indicator on the right is what changes between bars.

HOW A MODIFICATION IS RECOGNISED

The match is made against the Modified.Sequence string, ignoring case, and a class matches if either its UniMod identifier or its name appears in it:

- Carbamidomethylation: UniMod:4 or Carbamidomethyl, optionally followed by ation.
- Oxidation: UniMod:35 or Oxidation.
- Acetylation: UniMod:1 or Acetyl, optionally followed by ation.

The identifiers are matched so that a longer number cannot satisfy a shorter one:

UniMod:4 does not match UniMod:44, and UniMod:1 does not match UniMod:121. The names are matched so that they cannot be a fragment of a longer word.

The three classes are fixed. A modification outside them — a phosphorylation, an isobaric label, a pyroglutamate — is not counted here and does not appear in the figure, so this page describes the three it tracks and says nothing about the rest of the modifications in the report.

Edge cases

- A class with no positive count anywhere is dropped entirely: no bar, no legend entry, and the remaining bars widen to share the group.
- An annotated sample that matches no column in the report keeps its tick and its label and is drawn with no bars at all, so the gap is visible rather than silent.
- If no annotated sample matches any column, nothing is drawn and the page reports that the annotation does not match the report.
- A single replicate in a condition is drawn like any other sample; nothing on this page aggregates within a condition.
- A sequence whose cell is missing, non-numeric or exactly zero in a sample is not counted in that sample. It may still be counted in every other.
- A report without Modified.Sequence or without Precursor.Quantity draws nothing and names the missing column.

4 Exporting

The download reproduces the plot at the width, height and resolution set on the page. The browser scales the rendered figure to fit the page. Choose PDF for a figure that will be scaled or edited, PNG for a slide. The plot also appears in the summary report, which renders it from the settings left on this page.

Behind this page

Function	What it produces
modification_plot	Grouped bars of modified-sequence counts per sample, for the three tracked classes
peptide_modification_plot	The figure as the page renders it, including the appearance-panel settings. The counts behind the bars — one row per sample, one column per class, plus UniqueTrackedModifiedPeptides — are built by peptide_modification_counts.